

AI Academic Companion - Project Planning (Version 1)

1. Project Idea

A production-ready graduation project: an AI-powered academic platform for students, doctors, and administrators.

2. Project Scope

Version 1 includes authentication, AI chat, PDF upload, lecture summarization, quiz generation, course management, graduation project management, notifications, and dashboards.

3. Objectives

- Improve learning efficiency.
- Centralize academic services.
- Integrate AI into higher education.
- Provide scalable architecture.

4. User Roles

- Student
- Doctor
- Admin

5. Functional Requirements

- Register/Login
- Email verification
- AI assistant
- Upload files
- Summarize lectures
- Generate quizzes
- Notes
- Course management
- Graduation project management
- Notifications
- Dashboard

6. Non-Functional Requirements

- Security
- Performance
- Scalability
- Availability
- Maintainability
- Responsive UI

7. Tech Stack

Android: Kotlin, Jetpack Compose, MVVM, Hilt, Retrofit.

Backend: ASP.NET Core Web API, Clean Architecture, EF Core.

Database: PostgreSQL.

Authentication: Firebase + JWT.

AI: Gemini API.

Storage: Supabase Storage.

Dashboard: React.

8. System Architecture

Android/Web -> ASP.NET Core API -> PostgreSQL

-> Gemini API

-> Supabase Storage

-> Firebase Authentication

9. Database (Initial Modules)

Users, Roles, Departments, Courses, Assignments, Notes,
Files, AI Chats, Chat Messages, Quizzes, Projects, Notifications.

10. Modules

Authentication

AI Module

File Manager

Course Management

Study Planner

Notes

Quiz Generator

Graduation Project

Notifications

Dashboard

11. Development Roadmap

Planning -> System Design -> Database -> Backend -> Authentication -> Android -> AI
-> Dashboard -> Testing -> Deployment

12. Git Workflow

Commit after every completed feature using Conventional Commits.